

EXAMSPHERE

Software Requirements Specification (SRS)

Controller of Examination Management Portal

Simple and detailed requirements document for the proposed university examination automation system.

Document Item	Details
Project	ExamSphere — Controller of Examination Portal
Document	Software Requirements Specification
Version	1.0
Users	COE/Admin, Staff/Faculty, Examiners, Students, System Administrator
Purpose	Digitize and automate the complete examination lifecycle

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1. Introduction

ExamSphere is a web-based Controller of Examination Management Portal designed to digitize examination activities that are commonly handled through manual records, spreadsheets, paper forms and disconnected systems. It provides one platform for planning, registration, scheduling, hall allocation, attendance, valuation, results, certificates, revaluation, notifications and reports.

1.1 Purpose

- Define the functional and non-functional requirements of ExamSphere.
- Provide a common reference for developers, designers, testers and stakeholders.
- Describe what each major user can do.
- Provide a clear base for prototype development.

1.2 Scope

The system covers the examination lifecycle from examination creation and student registration through timetable publication, hall and seating allocation, attendance, valuation, result processing, revaluation and certificate delivery.

2. Problem Statement

Universities may face delays, repeated data entry, communication gaps, calculation errors and limited visibility when examination activities are managed manually or across separate systems. Confidential information such as question papers and marks also requires strict access control. ExamSphere solves these issues through one secure, role-based and workflow-driven platform.

3. Objectives

- Digitize manual examination processes.
- Provide role-based access.
- Automate eligibility, allocation and result calculations.
- Reduce data-entry and calculation errors.
- Provide real-time status and notifications.
- Generate reports and analytics.
- Maintain audit trails.
- Improve student and staff experience.

4. Stakeholders and User Roles

Role	Responsibilities
COE / Admin	Overall examination management, approvals, timetable, halls, results, reports and monitoring.
Staff / Faculty	Student verification, academic information, attendance, internal marks and assigned duties.
Examiner / Valuator	Question paper/valuation activities, script allocation and mark entry.
Student	Exam registration, fee status, timetable, hall ticket, results, revaluation and certificates.
System Administrator	User accounts, roles, security, configuration, integrations and maintenance.

5. Overall System Description

ExamSphere is a centralized web application. Users log in according to their roles, see the relevant dashboard and use permitted modules. Centralized data and connected workflows link the different stages of the examination process.

High-level flow: Academic Data → Exam Setup → Registration → Timetable → Centre/Hall Allocation → Seating → Exam Day → Valuation → Results → Revaluation/Certificates.

5.1 Main Modules

Module	Description
Dashboard	Shows examination status, KPIs, pending work and alerts.
Student Management	Manages student profile, eligibility and examination history.
Exam Management	Creates examinations and manages registration and eligibility.
Timetable	Creates, checks and publishes examination schedules.
Centre & Hall Management	Manages centres, halls and capacity.
Seating Arrangement	Allocates students to halls/seats and generates lists.
Staff & Examiner Management	Assigns invigilators and examiners.
Question Paper Management	Controls question paper setting, approval and secure access.
Attendance	Records presence/absence and incidents.
Valuation	Manages script allocation, valuation and mark entry.
Result Processing	Calculates grades/GPA/CGPA and prepares results.
Revaluation	Manages revaluation and retotalling.
Certificates	Provides mark sheets and digital certificates.
Notifications	Sends examination announcements and alerts.
Reports	Generates examination and performance reports.
Security & Administration	Manages roles, permissions, audit logs and settings.

6. Functional Requirements

ID	Requirement	Description
FR-01	Authentication	Users shall log in securely and be identified by role.
FR-02	Role-Based Access	Users shall see only permitted modules and actions.
FR-03	Student Management	Authorized staff shall search, view and manage student examination information.
FR-04	Exam Creation	COE/Admin shall create examination sessions and configure courses/rules.
FR-05	Exam Registration	Students shall view eligible subjects, register and view status.
FR-06	Eligibility	The system shall check required eligibility before registration.
FR-07	Fee Status	The system shall record and display examination fee/payment status.
FR-08	Timetable	Authorized users shall create, validate and publish timetables.
FR-09	Centre/Hall	The system shall maintain centres, halls and capacity.
FR-10	Seating	The system shall generate seating allocations using configured rules.
FR-11	Duty Allocation	Authorized users shall assign invigilators and examiners.
FR-12	Question Papers	Confidential question paper information shall be permission-controlled.
FR-13	Attendance	Authorized users shall mark attendance and generate absentee lists.
FR-14	Valuation	The system shall support script allocation, valuation and mark submission.
FR-15	Results	The system shall calculate grades and GPA/CGPA according to configured rules.
FR-16	Publication	Authorized COE users shall verify, approve and publish results.
FR-17	Student Results	Students shall view and download published results.
FR-18	Revaluation	Eligible students shall submit and track revaluation/retotalling requests.
FR-19	Certificates	Approved certificates/mark sheets shall be generated and accessed securely.
FR-20	Notifications	The system shall send relevant exam, timetable, hall and result notifications.

ID	Requirement	Description
FR-21	Reports	Authorized users shall generate examination and performance reports.
FR-22	Audit	Important actions shall be recorded in an audit trail.
FR-23	Search	Major modules shall support relevant search and filters.
FR-24	Downloads	Authorized users shall download permitted reports and documents.

7. User Interface / Page Requirements

Page	Required Content and Purpose
Login	Credentials, password recovery and optional MFA; secure entry to the correct role portal.
COE Dashboard	KPIs, upcoming exams, pending approvals, readiness, alerts and quick actions.
Student Management	Search, profile, programme, semester, eligibility, subjects and exam history.
Exam Registration	Eligible subjects, registration, fee status, submission and approval status.
Timetable	Date, time, subject, session, venue, conflict warning and publication controls.
Centre & Hall	Centres, halls, capacities, availability and allocation status.
Seating Arrangement	Student list, hall, seat number, allocation rules, layout and printable output.
Staff & Examiner	Availability, duty, hall/subject, workload and confirmation.
Question Paper	Subject, setter/moderator, approval status, secure access and release status.
Attendance	Hall, student list, present/absent, absentee list and incident reporting.
Valuation	Valuation centre, examiner, script allocation, marks, verification and submission.
Result Processing	Marks, grades, GPA/CGPA, pass/fail, arrears, exceptions and approval.
Student Results	Published results, subject grades, GPA/CGPA and downloads.
Revaluation	Eligible subjects, application, payment/status, updated marks and final result.
Certificates	Certificate list, request status, downloads and digital verification.
Notifications	Announcements, reminders, timetable/hall changes, result notifications and read status.
Reports	Registration, attendance, hall, valuation, result and performance reports with export.

8. Non-Functional Requirements

Area	Requirement
Security	Only authorized users shall access protected data. Sensitive information shall be protected and critical actions audited.
Performance	Common pages should load quickly under normal institutional use.
Availability	The portal should remain available during registration, examination and result periods.
Usability	Pages should be simple, consistent, responsive and easy to understand.
Scalability	The system should support growth in students, departments, programmes, examinations and campuses.
Reliability	Inputs should be validated and inconsistent examination data prevented.
Maintainability	Modules should be separated so future changes do not affect unrelated functions.
Auditability	Critical operations should record user, action and date/time.
Compatibility	The portal should work on modern desktop/tablet browsers.
Backup & Recovery	Important data should be backed up regularly and recoverable after failure.

9. Data Requirements

The centralized database should maintain:

- Users, roles and permissions.
- Student and academic information.
- Departments, programmes, semesters and courses.
- Examination sessions, registrations and fees.
- Timetables, centres, halls and seating.
- Staff, invigilator and examiner assignments.
- Question paper workflow metadata.
- Attendance, absentee and incident records.
- Valuation, scripts and marks.
- Grades, GPA/CGPA and published results.
- Revaluation requests and certificates.
- Notifications, reports and audit logs.

10. End-to-End Workflow

1. Student and academic data is available.
2. COE creates the examination and configures eligible courses.
3. Students register and complete required payment.
4. Staff/COE verify registration and eligibility.
5. COE prepares and publishes the timetable.
6. Centres and halls are configured and students are allocated.
7. Seating arrangements are generated.
8. Invigilators and examiners receive assignments.
9. Attendance is recorded.
10. Answer scripts move through valuation and mark entry.
11. Results calculate grades and GPA/CGPA.
12. COE reviews and approves results.
13. Results are published and students are notified.
14. Eligible students request revaluation/retotalling.
15. Approved mark sheets and certificates are made available.

11. External System Integration

System	Purpose
Student Information System / ERP	Synchronize student, programme and academic information.
Institutional Identity / SSO	Centralized authentication where available.
Payment Gateway	Process examination fees and return payment status.
Email / SMS Service	Send notifications and alerts.
Finance System	Support fee reconciliation and financial records.
Digital Verification Service	Support QR or online certificate verification.

12. Security Requirements

- Use role-based access control.
- Protect passwords using secure authentication mechanisms.
- Use MFA for privileged accounts where appropriate.
- Encrypt sensitive data during transmission and protect confidential documents.
- Restrict question paper, marks and result access to authorized roles.
- Maintain audit logs for critical actions.
- Use secure sessions and appropriate session expiry.
- Back up examination data regularly.
- Follow institutional privacy and data-retention policies.

13. Reports and Analytics

The portal should provide simple dashboards and downloadable reports for monitoring and decision-making.

- Student registration report.
- Eligibility and arrear report.
- Timetable and hall allocation report.
- Seating arrangement report.
- Attendance and absentee report.
- Invigilator/examiner duty report.
- Valuation completion report.
- Pass/fail and grade distribution report.
- Department and subject performance report.
- Revaluation report.
- Certificate/report generation status.

14. Assumptions and Constraints

- Users have valid institutional accounts or credentials.
- Student and academic master data is available from the institution.
- Eligibility, grading and attendance rules are configurable.
- Payment and notification services may depend on external providers.
- Exact examination policies vary by institution and must be configurable.
- The prototype may implement a limited set of workflows while supporting future modules.

15. Basic Acceptance Criteria

The prototype is successful when it demonstrates:

- Role-based login and dashboards.
- COE examination creation and monitoring.
- Student registration for eligible subjects.
- Timetable creation and publication.
- Hall and seating allocation.
- Attendance recording.
- Mark entry and sample result processing.
- Student result viewing.

- Notifications.
- Report generation.
- Permission protection and audit tracking.

16. Future Enhancements

- AI-based exam readiness prediction.
- Smart hall and seating optimization.
- Workload balancing for invigilators and examiners.
- Anomaly detection in marks and examination operations.
- Advanced student performance analytics.
- Mobile application.
- Digital signatures for certificates.
- Multi-campus and multi-university support.

17. Conclusion

ExamSphere provides a centralized and secure solution for the complete Controller of Examination workflow. By digitizing manual activities, automating repetitive processes, providing role-based access, enabling real-time communication and generating useful reports, the system can improve efficiency, transparency and user experience.

This SRS provides a clear foundation for designing the UI prototype, database, backend services and future production implementation.